Sparky - wiring, assembly and test guide

Circuit v3, for the parts that arrived from Communica, Micro Robotics and DIY Electronics. Build it on the desk first, test each part as you add it, then move it into Sparky.

What changed since the report's wiring diagram (v2):

- **Accent LED moved to 3.3 V** - the report's identified risk. It now runs 3.3 V rail -> 47 Ω -> LED -> GPIO 4, so GPIO 4 never sees more than 3.3 V. Still lit by GPIO 4 LOW.
- **The ESP32 sits off the breadboard**, on female-to-male jumpers. A 38-pin ESP32 on a half breadboard covers every free hole next to one of its rows of pins.
- **Amp GAIN is optional**: open = 9 dB to start, to GND = 12 dB if Sparky is too quiet.
- **The LED is a plain white 5 mm LED** (what was bought), not a WS2812B.

You still need (not in the orders):

Item	How many	Why / where to get it
Female-to-male jumper wires, 20 cm	13 (get a 40-pack)	ESP32 to breadboard. The M-F line was refunded - borrow from Mr Blackie or the lab, or any electronics shop
Resistor 47 Ω (33-68 Ω is fine)	1	Accent LED. Colour bands yellow-violet-black. Lab drawer
Soldering iron + solder	-	Mic header pins (they come loose). Amp too, if its pins are loose
Multimeter	-	Power checks before switching on (section 4)
Micro-USB data cable	1	A charge-only cable powers the ESP32 but can't upload

Three rules that protect the parts: (1) the microphone gets 3.3 V only - 5 V destroys it; (2) never connect 5 V to a GPIO pin or to the ESP32's 3V3 pin; (3) unplug the USB cable whenever you move a wire.

1. Pinout - what connects to what

Sparky - circuit pinout (v3)

ESP32-DevKitC 38-pin seen from the top (antenna up, USB down). Coloured tags = pins Sparky uses. Go by the labels printed on your boards.

INMP441 microphone

VDD	→ 3.3 V rail
GND	→ GND
L/R	→ GND (left channel)
SCK	→ GPIO 32
WS	→ GPIO 25
SD	→ GPIO 33

3.3 V ONLY - 5 V destroys the mic

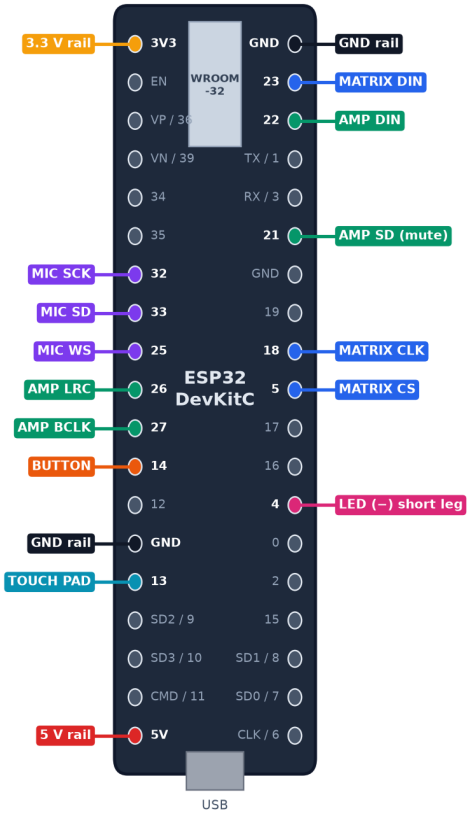
MAX98357A amp + 3 W speaker

VIN	→ 5 V rail
GND	→ GND
LRC	→ GPIO 26
BCLK	→ GPIO 27
DIN	→ GPIO 22
SD	→ GPIO 21 (LOW = mute)
GAIN	→ open = 9 dB, to GND = 12 dB
+ / −	→ speaker red / black

Push-to-talk button

leg 1	→ GPIO 14 (internal pull-up)
leg 2	→ GND

Tactile: use DIAGONAL legs. Arcade: COM + NO



MAX7219 LED matrix 32 x 8 (IN end)

VCC	→ 5 V rail
GND	→ GND
DIN	→ GPIO 23
CS	→ GPIO 5
CLK	→ GPIO 18

Use the IN end (DIN, not DOUT). Keep brightness low on USB.

Accent LED (white 5 mm, now on 3.3 V)

3.3 V rail	→ 47 Ω resistor
resistor	→ LED long leg (+)
short leg (−)	→ GPIO 4 (LOW = on)

Touch pad

pad / wire	→ GPIO 13 (touch T4)

Bench: bare jumper end. In Sparky: copper tape pad

Power (USB 5 V into the ESP32)

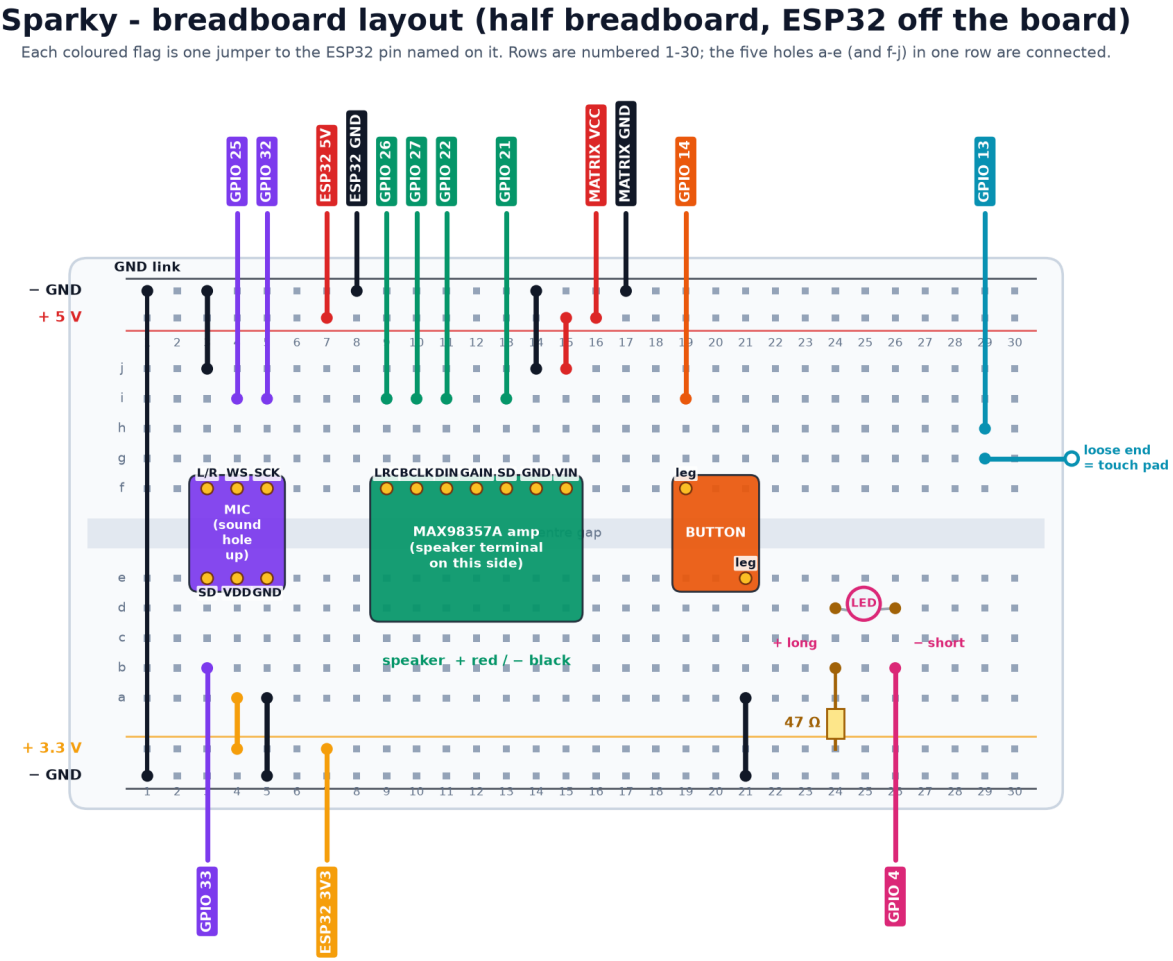
ESP32 5V	→ top + rail = 5 V
ESP32 3V3	→ bottom + rail = 3.3 V
ESP32 GND	→ both − rails (link them)

Never feed 5 V into any GPIO or the 3V3 pin

5 V · 3.3 V · GND · mic · amp · matrix · button · LED · touch

Tag colours match the wire colours used on the next page and in the checklist. Grey pins are unused - leave them free (GPIO 0, 2, 12 and 15 also affect start-up).

2. Breadboard layout



Matrix DIN / CS / CLK go straight from its cable to ESP32 GPIO 23 / 5 / 18 (no breadboard). Pin order on your mic and amp may differ - wire by the printed label, not by position.

- **Rails:** top + = 5 V (matrix, amp), bottom + = 3.3 V (mic, LED). Both – rails are GND - link them with one jumper at the end. Check your breadboard: some rails are split in the middle; if so, bridge the two halves too.
- **Mic** straddles the centre gap like a chip - its two rows of pins must be on opposite sides. Both rows on the same side shorts the pins together. Sound hole facing up.
- **Amp** pins go into column f, rows 9-15, with the board lying over the gap. Connect through the free holes g-j in the same rows.
- **Button:** a tactile switch across the gap, wired by *diagonally opposite* legs - that always works, whichever way round it is. The arcade button replaces it later (its COM and NO terminals take the same two wires).

3. Wire checklist

Work top to bottom and tick each wire. The order matches section 4: power first, then one part at a time, each tested before the next.

#	Part	From	To	Wire	Done
1	Power	ESP32 5V	Top + rail (5 V)	F-M jumper	[ ]
2	Power	ESP32 3V3	Bottom + rail (3.3 V)	F-M jumper	[ ]
3	Power	ESP32 GND (either)	Top – rail (GND)	F-M jumper	[ ]
4	Power	Top – rail (GND)	Bottom – rail (GND), at row 1	M-M jumper	[ ]
5	Matrix	Matrix VCC (IN end)	Top + rail (5 V)	Matrix cable + M-M pin	[ ]
6	Matrix	Matrix GND	Top – rail (GND)	Matrix cable + M-M pin	[ ]
7	Matrix	Matrix DIN	ESP32 GPIO 23	Matrix cable (F-F)	[ ]
8	Matrix	Matrix CS	ESP32 GPIO 5	Matrix cable (F-F)	[ ]
9	Matrix	Matrix CLK	ESP32 GPIO 18	Matrix cable (F-F)	[ ]
10	Amp	Amp VIN (row 15)	Top + rail (5 V)	M-M jumper	[ ]
11	Amp	Amp GND (row 14)	Top – rail (GND)	M-M jumper	[ ]
12	Amp	Amp LRC (row 9)	ESP32 GPIO 26	F-M jumper	[ ]
13	Amp	Amp BCLK (row 10)	ESP32 GPIO 27	F-M jumper	[ ]
14	Amp	Amp DIN (row 11)	ESP32 GPIO 22	F-M jumper	[ ]
15	Amp	Amp SD (row 13)	ESP32 GPIO 21	F-M jumper	[ ]
16	Amp	Amp GAIN (row 12)	Nothing at first (9 dB). Louder: to GND (12 dB)	-	[ ]
17	Amp	Speaker red / black	Amp screw terminal + / –	Speaker's own wires	[ ]
18	Mic	Mic VDD (row 4, bottom)	Bottom + rail (3.3 V) - NEVER 5 V	M-M jumper	[ ]
19	Mic	Mic GND (row 5, bottom)	Bottom – rail (GND)	M-M jumper	[ ]
20	Mic	Mic L/R (row 3, top)	Top – rail (GND) = left channel	M-M jumper	[ ]
21	Mic	Mic SD (row 3, bottom)	ESP32 GPIO 33	F-M jumper	[ ]
22	Mic	Mic WS (row 4, top)	ESP32 GPIO 25	F-M jumper	[ ]
23	Mic	Mic SCK (row 5, top)	ESP32 GPIO 32	F-M jumper	[ ]
24	Button	Button leg, row 19 top	ESP32 GPIO 14	F-M jumper	[ ]
25	Button	Button leg, row 21 bottom (diagonal)	Bottom – rail (GND)	M-M jumper	[ ]
26	LED	47 Ω resistor	Bottom + rail (3.3 V) to row 24	Resistor legs	[ ]
27	LED	LED long leg (+)	Row 24 (with the resistor)	LED leg	[ ]
28	LED	LED short leg (–, flat side)	Row 26	LED leg	[ ]
29	LED	Row 26	ESP32 GPIO 4	F-M jumper	[ ]
30	Touch	Row 29 top	ESP32 GPIO 13	F-M jumper	[ ]
31	Touch	Row 29 top	Loose jumper end = touch pad	M-M jumper	[ ]

4. Assembly and test - one part at a time

Upload the test program once, before wiring anything: it then runs every time the ESP32 powers up. It is `firmware/sparky_hw_test` in the repository - ask Claude to upload it, or:

```
arduino-cli upload --fqbn esp32:esp32:esp32 -p COM5 firmware/sparky_hw_test
arduino-cli monitor -p COM5 -c baudrate=115200
```

In the monitor, type the test number and press Enter. **q** stops a long test. Each test prints what a pass looks like. Test IDs (M-xx) are the report's module tests.

Step A - power only

1. Wire the four power rows of the checklist (ESP32 5V, 3V3, GND, GND link). Nothing else yet.
2. **USB unplugged**, multimeter on continuity (beep): top + to top – must **not** beep; bottom + to bottom – must **not** beep; top + to bottom + must **not** beep.
3. Plug in USB. Volts mode: top + to – = **4.6-5.1 V**; bottom + to – = **3.2-3.4 V**. Unplug.

Step B - LED matrix (M-05) -> test 1

Wire the 5 matrix rows. Test 1 first lights a single dot: it must be in the **top-left** corner seen from the front. If it's top-right, set `SPARKY_FLIP_X 0`; bottom-left, set `SPARKY_FLIP_Y 1` - in both `sparky_hw_test.ino` and `firmware/sparky/config.h`. Then the six mood faces show for 2.5 s each.

Step C - amp and speaker (M-04) -> test 2

Wire the amp rows and the speaker. Low beep, high beep, then silence. Hiss in the silence means the mute (SD, GPIO 21) isn't connected.

Step D - microphone (M-01, M-02) -> tests 3, 4, 5

Before plugging in USB: with the multimeter on continuity, mic VDD to the bottom + rail (3.3 V) **beeps**, and mic VDD to the top + rail (5 V) does **not**. Test 3 shows a level bar for each channel: one should jump when you talk. Test 4 records 2.5 s and plays it back - you should hear yourself. Test 5 counts samples for 10 s: about 16 000 per second.

Step E - button (M-07) -> test 6

Press 100 times; the count must end at exactly 100. It counts by itself without pressing = the button is wired to the wrong pair of legs (use diagonal legs).

Step F - accent LED and touch pad (M-20) -> tests 8 and 7

Test 8 blinks three times, then pulses. Test 7 calibrates for 2 s (don't touch), then counts 50 taps on the loose jumper end.

Step G - power under load (M-08) -> test 9

All 256 LEDs on with a beep at brightness 2, 5 and 8. If the ESP32 restarts, the top line says BROWNOUT: use a 5 V / 2 A phone charger instead of a laptop port, and keep the face brightness low. A USB power meter, if you have one, gives the M-08 numbers.

5. If something doesn't work

Symptom	Most likely cause
Nothing at all, no USB port appears	Charge-only cable; try another cable/port. Windows may need the CP210x or CH340 USB driver
Upload fails 'waiting for download'	Hold the BOOT button while the upload starts
Matrix blank	Cable on the DOUT end instead of DIN; VCC/GND missing; CS on the wrong pin
Matrix shows garbage / random dots	CLK and DIN swapped; loose GND
Faces mirrored or upside down	SPARKY_FLIP_X / SPARKY_FLIP_Y (step B)
No sound	Amp VIN not on 5 V; SD not on GPIO 21; speaker not tight in the terminal
Constant hiss	SD (mute) not connected - the amp is never muted
Distorted or too loud	Leave GAIN open (9 dB); move the speaker away from the mic
Mic level always 0	VDD not on 3.3 V; SCK / WS / SD swapped; mic not straddling the gap
Mic level always full	GND missing; L/R floating (tie it to GND)
Recording sounds fast or slow	Test 5 reads far from 16 000: check SCK and WS
Button counts by itself	Wrong legs - use diagonal legs; GPIO 14 wire loose
LED never lights	LED backwards (long leg goes to the resistor); resistor in the wrong row
Touch counts without touching	Loose wire too long or near the matrix cable - shorten it
Restarts when the matrix lights	Brownout - stronger 5 V supply, lower brightness

6. When every test passes

1 Upload the real Sparky program: `firmware/sparky` (same command, different folder). Its README lists nine more checks: Wi-Fi setup, Studio sync, live preview, voice lines, power cut.

2 Move into Sparky: same connections, longer F-M jumpers to the mic (forehead) and amp (belly). Swap the tactile switch for the arcade button and the loose touch wire for a copper-tape pad.

3 Re-run tests 1, 2, 4 and 6 inside the enclosure - wires get bumped during assembly.

Results sheet for the report

Test	Part	Pass criterion (report Table)	Result / date
M-01	Microphone	Recording audible, no clipping or dropouts, 10 of 10	
M-02	Mic bus	Sample rate as configured (16 000 Hz), no frame slip	
M-04	Speaker	Intelligible at 2 m, no distortion	
M-05	Face display	Every face distinct and recognisable at arm's length	
M-07	Button	100 presses = 100 events	
M-08	Power	No brownout under peak load (charger and power bank)	
M-20	LED, touch	Patterns distinct; 50 taps = 50 touches	